

Titration report



Method

Method ID	SERCALKSAMPLES	Release state	Released
Name	TA of Seawater	Modified by	SystemInternal
Modified on	2/6/2024 11:26:27 AM		

Summary

Samples

Sample ID	Name	Result	State
1	CRM 209-0498, 1	Sample size Content R1	57.694 g 2.2133 mmol/L OK

Main Result Details

Result title	Content	Sample ID1	CRM 209-0498
Result name	R1	Sample ID2	1
Result	2.2133 mmol/L	Result comment	
		Result state	OK
		Result state comment	
		Nominal value	
		Compliance state	Released
		Is result deleted	No

Calculation formula R1=Q*C/m

Raw Results

Scope 1/1, Sample 1/1 (CRM 209-0498; 1)

Sample (S1)

Creation time	2/22/2024 10:35:14 AM
Sample start time	2/22/2024 10:35:14 AM
Sample size	57.694 g
Correction factor	1.0
Temperature	25.0 °C
Density	1.022 g/mL

Titration stand (Titrationstand1)

Creation time	2/22/2024 10:35:14 AM
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Measure (normal) [1] (Measurenormal1)

Creation time	2/22/2024 10:35:14 AM
T	23.0 °C
E	7.919 pH
tMe	01:15 min
ESTMe	2.952 pH

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Titration (EQP) [1] (TitrationEQP1)

Creation time 2/22/2024 10:36:30 AM
Terminated at Potential
EEQ1 4.441 pH ⁽⁹⁾
EHN1 6.106 pH
Q1 0.127692 mmol ⁽⁹⁾
VEQ1 1.275645 mL ⁽⁹⁾
EST 7.916 pH
VEX 0.783605 mL
VEND 2.05925 mL
t 23:22 min
QEND 0.206131 mmol
QEX 0.078439 mmol
EPD 6.647 pH
nEQ 1

Content (R1)

Creation time 2/22/2024 10:59:57 AM
Result 2.2133 mmol/L
Decimal places 4
Substance None
M 1 g/mol
z 1

Rinse (Rinse1)

Creation time 2/22/2024 10:59:58 AM

After Sample Scopes

Park (Park1)

Creation time 2/22/2024 11:00:10 AM

Measure (normal) [1] Scope 1/1, Sample 1/1

Measured Values Scope 1/1, Sample 1/1

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	2.952	17.7	
2	4.766	18.2	1.814
3	5.327	18.9	0.561
4	5.684	19.7	0.357
5	5.982	20.3	0.298
6	6.243	20.8	0.261
7	6.479	21.3	0.236
8	6.666	21.6	0.187
9	6.848	21.8	0.182
10	7.004	22.0	0.156
11	7.139	22.2	0.135
12	7.254	22.3	0.115

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13	7.357	22.4	0.103
14	7.438	22.5	0.081
15	7.515	22.6	0.077
16	7.579	22.6	0.064
17	7.634	22.7	0.055
18	7.681	22.7	0.047
19	7.724	22.7	0.043
20	7.755	22.7	0.031
21	7.780	22.8	0.025
22	7.803	22.8	0.023
23	7.828	22.8	0.025
24	7.846	22.8	0.018
25	7.862	22.8	0.016
26	7.874	22.8	0.012
27	7.884	22.8	0.010
28	7.892	22.8	0.008
29	7.900	22.8	0.008
30	7.906	22.9	0.006
31	7.911	22.9	0.005
32	7.915	22.9	0.004
33	7.919	22.9	0.004
34	7.921	22.9	0.002
35	7.923	22.9	0.002
36	7.925	22.9	0.002
37	7.926	22.9	0.001
38	7.927	22.9	0.001
39	7.928	22.9	0.001
40	7.928	22.9	0.000
41	7.929	22.9	0.001
42	7.929	22.9	0.000
43	7.929	22.9	0.000
44	7.929	22.9	0.000
45	7.928	22.9	-0.001
46	7.928	22.9	0.000
47	7.927	22.9	-0.001
48	7.927	23.0	0.000
49	7.926	23.0	-0.001
50	7.926	23.0	0.000
51	7.925	23.0	-0.001
52	7.924	23.0	-0.001
53	7.924	23.0	0.000
54	7.923	23.0	-0.001
55	7.923	23.0	0.000
56	7.922	23.0	-0.001
57	7.921	23.0	-0.001
58	7.920	23.0	-0.001
59	7.919	23.0	-0.001
60	7.919	23.0	0.000

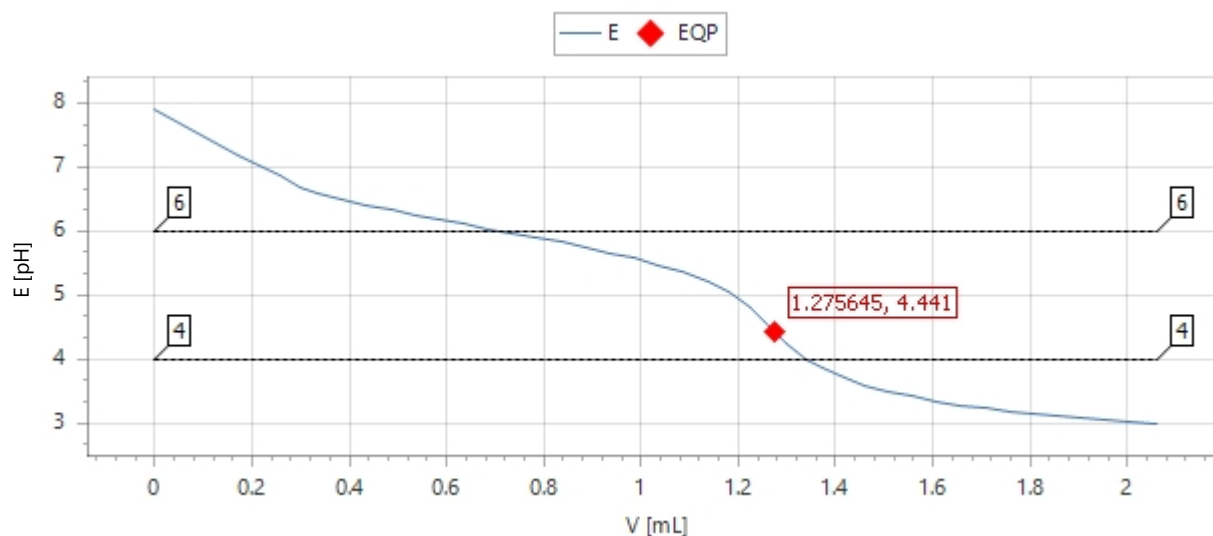
Titration (EQP) [1]

Scope 1/1, Sample 1/1

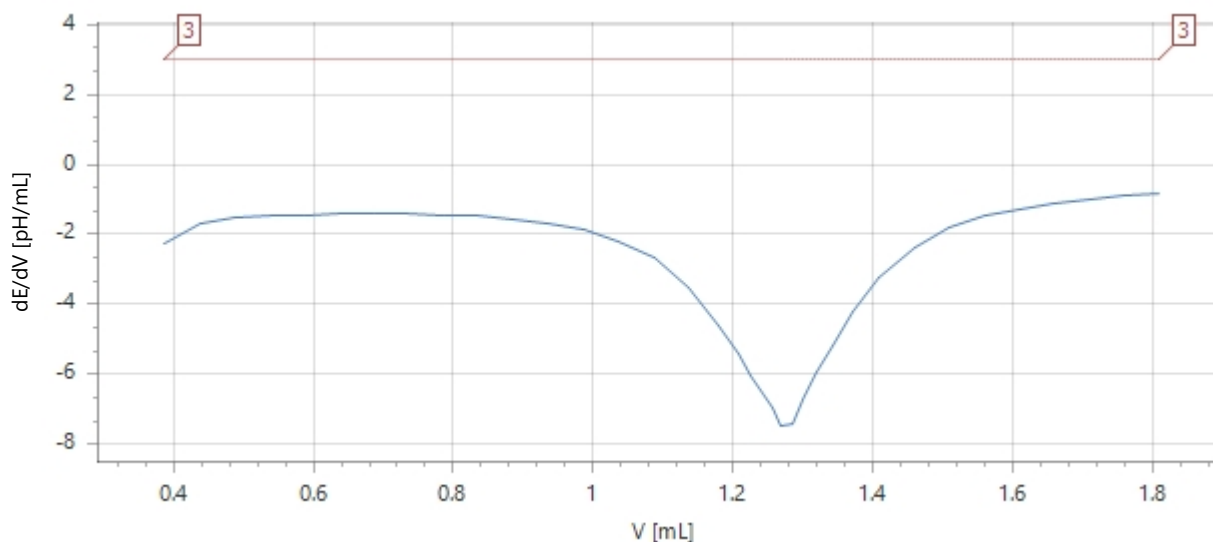
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E - V Curve Scope 1/1, Sample 1/1



dE/dV - V Curve Scope 1/1, Sample 1/1



Measured Values Scope 1/1, Sample 1/1

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.916		23.0			
28	0.17150	7.189	0.17150	23.2	-0.727		
59	0.25725	6.858	0.08575	23.3	-0.331		
89	0.30000	6.697	0.04275	23.4	-0.161		
180	0.33675	6.587	0.03675	23.7	-0.110		
210	0.38675	6.500	0.05000	23.8	-0.087	-2.25	
240	0.43675	6.410	0.05000	23.9	-0.090	-1.70	
270	0.48675	6.331	0.05000	24.0	-0.079	-1.49	
301	0.53675	6.250	0.05000	24.0	-0.081	-1.46	
331	0.58675	6.179	0.05000	24.1	-0.071	-1.45	
361	0.63675	6.108	0.05000	24.2	-0.071	-1.41	
391	0.68675	6.038	0.05000	24.2	-0.070	-1.39	

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422	0.73675	5.969	0.05000	24.3	-0.069	-1.40	EQP
452	0.78675	5.894	0.05000	24.3	-0.075	-1.43	
482	0.83675	5.824	0.05000	24.4	-0.070	-1.48	
512	0.88675	5.749	0.05000	24.4	-0.075	-1.56	
543	0.93675	5.666	0.05000	24.5	-0.083	-1.67	
573	0.98675	5.575	0.05000	24.5	-0.091	-1.86	
603	1.03675	5.476	0.05000	24.5	-0.099	-2.21	
633	1.08675	5.357	0.05000	24.6	-0.119	-2.68	
664	1.13675	5.212	0.05000	24.6	-0.145	-3.52	
694	1.18050	5.045	0.04375	24.6	-0.167	-4.60	
724	1.20750	4.903	0.02700	24.6	-0.142	-5.41	
755	1.22625	4.812	0.01875	24.7	-0.091	-6.03	
785	1.25575	4.594	0.02950	24.7	-0.218	-6.94	
815	1.26800	4.505	0.01225	24.7	-0.089	-7.46	
	1.275645 ⁽⁹⁾	4.441 ⁽⁹⁾					
846	1.28675	4.347	0.01875	24.7	-0.158	-7.44	
876	1.30075	4.240	0.01400	24.8	-0.107	-6.73	
906	1.32025	4.106	0.01950	24.8	-0.134	-5.95	
936	1.34225	3.983	0.02200	24.8	-0.123	-5.18	
966	1.37200	3.849	0.02975	24.8	-0.134	-4.20	
997	1.40925	3.722	0.03725	24.8	-0.127	-3.23	
1027	1.45925	3.593	0.05000	24.9	-0.129	-2.35	
1057	1.50925	3.493	0.05000	24.9	-0.100	-1.82	
1088	1.55925	3.411	0.05000	24.9	-0.082	-1.48	
1118	1.60925	3.342	0.05000	24.9	-0.069	-1.26	
1148	1.65925	3.282	0.05000	24.9	-0.060	-1.10	
1178	1.70925	3.230	0.05000	24.9	-0.052	-0.98	
1208	1.75925	3.183	0.05000	24.9	-0.047	-0.88	
1239	1.80925	3.142	0.05000	25.0	-0.041	-0.80	
1269	1.85925	3.103	0.05000	25.0	-0.039		
1299	1.90925	3.067	0.05000	25.0	-0.036		
1330	1.95925	3.035	0.05000	25.0	-0.032		
1360	2.00925	3.005	0.05000	25.0	-0.030		
1390	2.05925	2.977	0.05000	25.0	-0.028		

Used Instruments

Name	SERCtitrate	Serial number	B433917554
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Responsible User

Time	Login name	Full name
2/22/2024 10:35:12 AM	SystemInternal	System internal account

(9) Standard evaluation used